

Abstract No.

Dissecting the contribution of ZBTB18 and TIP60 complex to the mesenchymal/quiescent state in GBM

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The mesenchymal-like (MES-like) cell state of glioblastoma (GBM) has been associated with its most aggressive features. Single-cell RNAseq studies have revealed that MES-like tumors contain a high proportion of slow-cycling cells, which are responsible for therapy resistance, tumor recurrence and the establishment of the GBM immunosuppressive environment. Interestingly, upon recurrence, a mesenchymal-like shift often occurs. It seems this is, at least partly, caused by quiescent mesenchymal cells (qGBM) that have increased their proliferation rate or exited quiescence. Importantly, qGBM cells also evade immune surveillance, contributing to the establishment of the GBM immunosuppressive environment.

We have previously pinpointed Zinc finger and BTB domain-containing protein 18 (ZBTB18) as a transcriptional repressor of mesenchymal genes and a tumor suppressor in GBM. In order to get more insight into ZBTB18-mediated transcriptional repression mechanism, we performed co-immunoprecipitation experiments, followed by mass spectrometry (MS), in brain tumor stem-like cells (BTSC), which express ZBTB18. The analysis revealed that many components of the NuA4/TIP60 acetyltransferase complex are putative ZBTB18 interactors. The catalytic subunit KAT5 of this complex has been recently shown to oppose the quiescent state in GBM. Based on previous reports and on our preliminary data, we hypothesize ZBTB18 and NuA4/TIP60 complex interact to regulate the expression of genes important for the mesenchymal/quiescent GBM state. Examination of the acetylation status of histone H4 by ChIPseq in BTSCs revealed a global enrichment of acetylated peaks at genes previously linked to qGBM cells upon both ZBTB18 and KAT5 ko. In particular, our data suggest that ZBTB18 and NuA4/TIP60 complex may inhibit the MES/qGBM state by interfering with TGFB3-mediated gene activation.



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